

A Multi-region Logistics SaaS on NATS

How shippers and transporters meet through a governed marketplace; how NATS accounts isolate tenants; and how the same service and data model deploys across regions and logistics edge locations.

THE GOVERNING INVARIANT

Region is placement. Tenant is authority. Organisation is participation. A marketplace tenant may span regions and contain many shippers and transporters, but every interaction is mediated by platform services and constrained by account, membership, role, and subject permissions.

WHAT THE PUBLIC LINEBOOKER MODEL ADDS

The business loop is not merely "customer talks to transporter." It is **procure, aggregate, execute and settle**: fixed-lane tenders, bidding loads, truck commitment, route execution, live tracking, digital POD, claims and consolidated payment.

THE TEN ARCHITECTURE CARDS

- 01 · System landscape
- 02 · Shipper-transporter engagement
- 03 · Service and bounded-context model
- 04 · Tenant, organisation and account model
- 05 · Human identity to NATS session
- 06 · Regional deployment model
- 07 · Data and event ownership
- 08 · Tracking and logistics edge
- 09 · Failure, recovery and residency
- 10 · Provisioning and operations

01

REGION IS PLACEMENT · ACCOUNT IS AUTHORITY

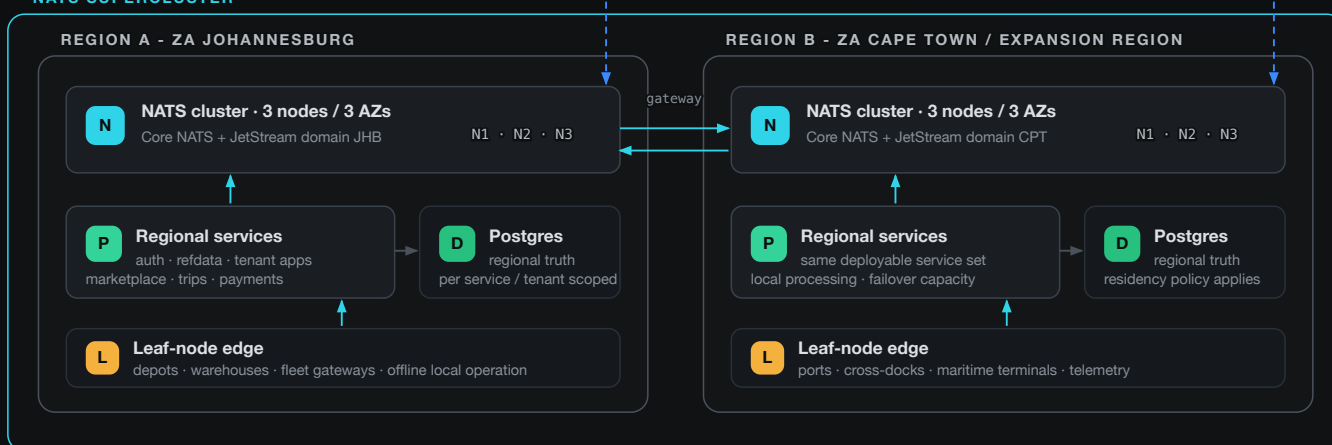
One SaaS, many regions, hard tenant walls

A regional NATS supercluster carries the live logistics fabric. Human identity enters through WorkOS; signed NATS credentials place every browser and service inside exactly one account boundary.

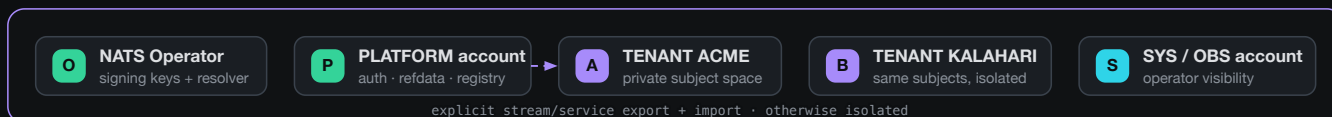
GLOBAL ACCESS AND IDENTITY PLANE



NATS SUPERCLUSTER



LOGICAL ACCOUNT PLANE - RESOLVED AND ENFORCED BY EVERY REGIONAL NATS SERVER



Solid arrow = request/read · Dashed arrow = pushed credential/event · Dotted = selective replication · Blue = client · Cyan = NATS · Green = platform · Purple = tenant

GEOGRAPHY

Clusters provide regional HA.

Gateways make the clusters one supercluster. A tenant can operate in several regions without region becoming part of its identity.

IDENTITY + TENANCY

WorkOS identifies the human; NATS authorizes the connection.

The minted credential binds a user or service to one account and a least-privilege subject set.

DATA PLACEMENT

Persistence stays regional by policy.

Postgres is authoritative; JetStream domains retain local event history. Mirror only the streams approved for DR or global views.

SUBJECT
TAXONOMY

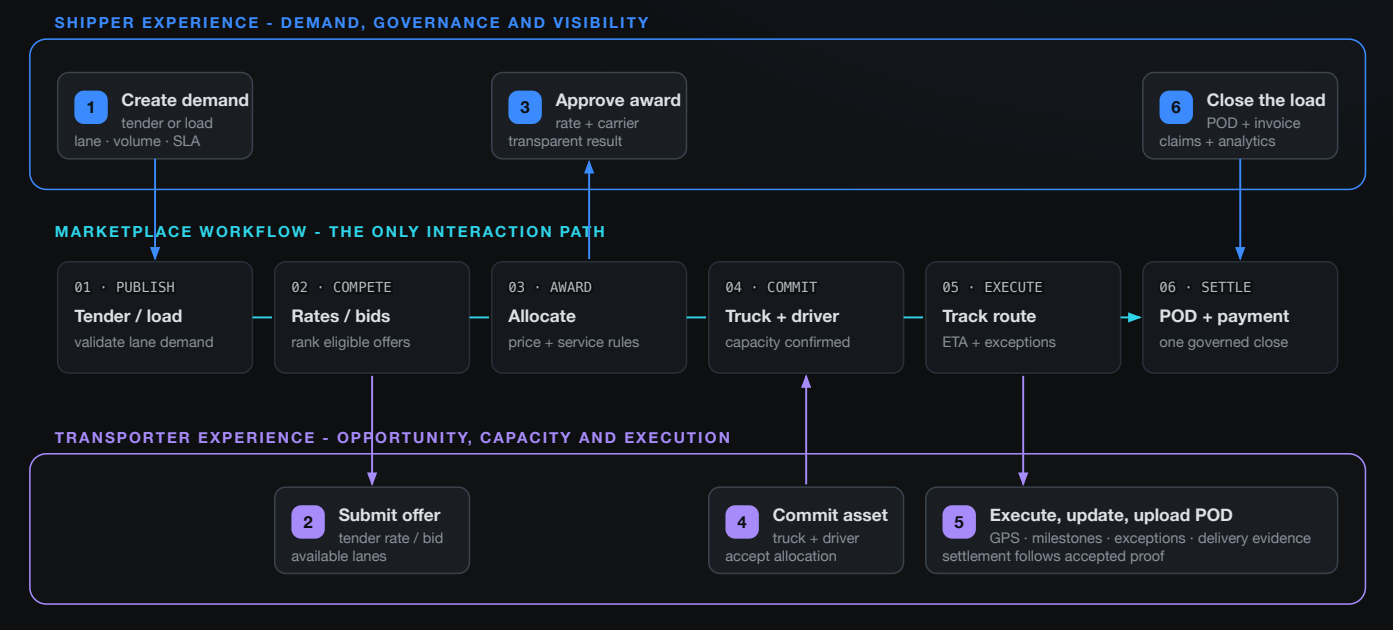
Four subject families carry every interaction. `evt.{context}.{service}.{entity}.{entity-id}.{event}` for replayable domain facts, `rpc.*` between services, `api.*` from a browser and `notify.*` for live change. `{context}` is the business unit — never the tenant and never the region, because tenancy is the account wall and region is a deployment choice.

02

THE MARKETPLACE IS THE CONVERSATION

How shippers and transporters engage

They never receive a direct technical channel to one another. Each side acts through a role-specific client; marketplace services validate, rank, allocate, notify and preserve the audit trail.



Blue = shipper action · Purple = transporter action · Cyan = marketplace state transition · Every transition is authorised, persisted and observable

FIXED CONTRACT LANES

Longer-term tenders establish lane rates, committed volumes and selected transporter capacity.

DAILY BIDDING LOADS

Spot opportunities absorb excess fleet and return legs without bypassing vetting or allocation rules.

EXECUTION CLOSES THE LOOP

Tracking, exceptions, digital POD, claims and settlement remain part of the same governed load record.

ENGAGEMENT RIGHT	SHIPPER CLIENT	MARKETPLACE SERVICE	TRANSPORTER CLIENT
Publish tender or load	Yes, for its own organisation	Validates lane, volume and SLA	Reads eligible opportunities only
Submit rate or bid	Not permitted	Ranks offers on price and service	Yes, on vetted lanes and truck types
Award and allocate	Approves the presented result	Applies allocation rules and records the decision	Accepts or declines the allocation
Update trip and tracking	Reads the permitted window	Normalises fixes and keeps provenance	Publishes position and milestones
Prove and settle	Accepts or disputes the evidence	Gates settlement on accepted proof	Uploads delivery evidence

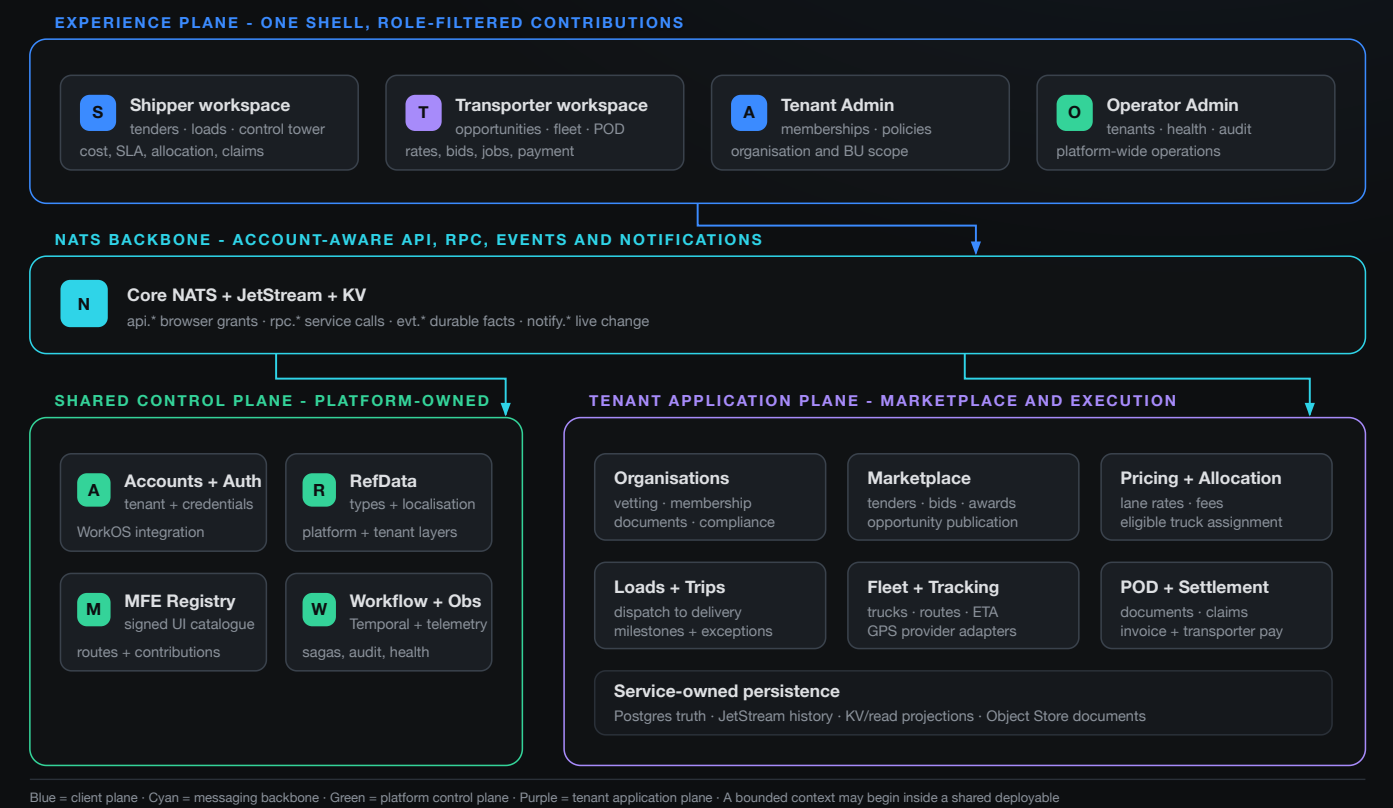
VETTING GATE **Capacity is admitted before it can compete.** A transporter is registered, vetted and onboarded before any tender or bidding subject is granted, so allocation draws only from approved carriers and agreed truck types.

03

CONTROL PLANE AROUND AN APPLICATION PLANE

Services follow business ownership

The boxes below are bounded contexts and deployable ownership units. They are not a mandate to create one container, database or team for every noun on day one.



DEPLOYMENT
RULE

Package around change cadence and ownership. Split a bounded context into its own service only when independent scaling, release, failure isolation or data ownership justifies the operational cost.

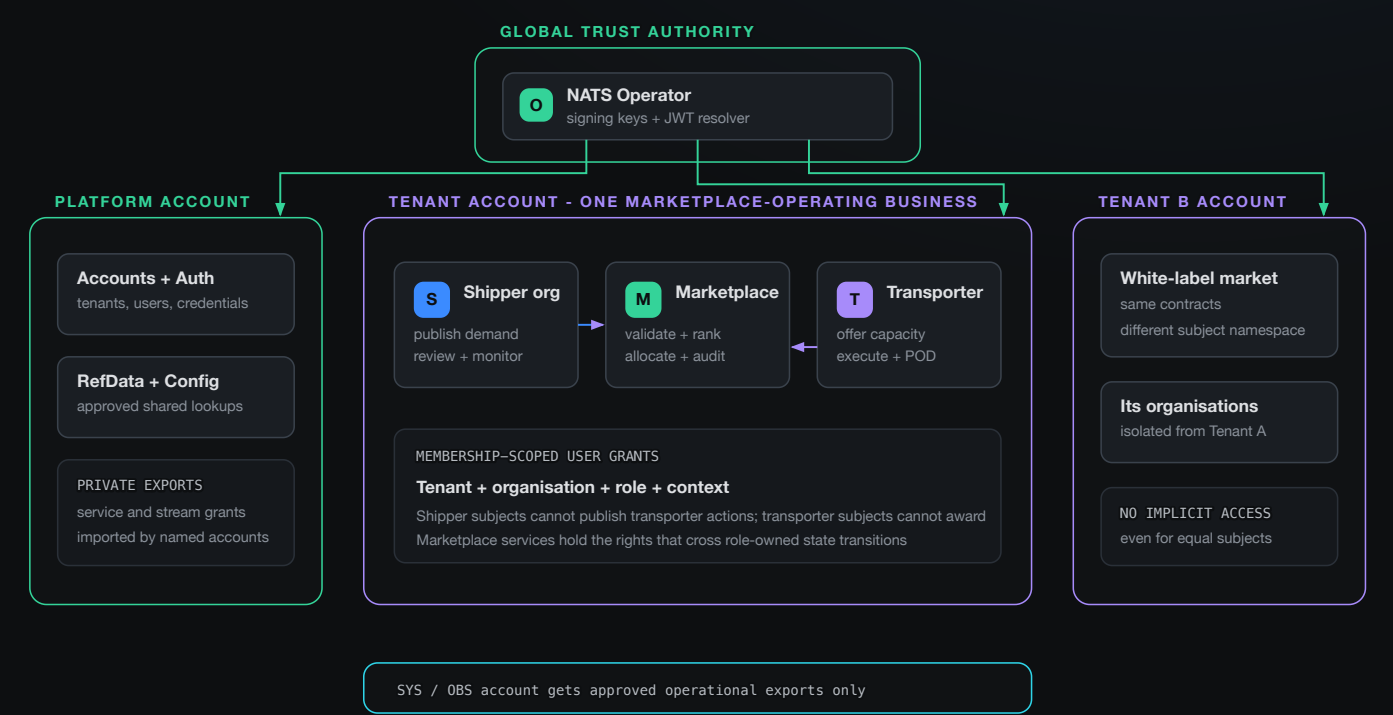
BOUNDED CONTEXT	OWNS	PUBLISHES	CONSUMES
Organisations	Vetting, membership, documents, compliance	Organisation and membership facts	Auth memberships, reference data
Marketplace	Tenders, bids, awards	Award and allocation facts	Organisation eligibility, lane pricing
Loads + Trips	Dispatch, route plan, milestones, exceptions	Trip lifecycle facts	Allocation, tracking, fleet state
Fleet + Tracking	Trucks, drivers, normalised position	Position and exception facts	Provider adapters, trip plan
POD + Settlement	Documents, charges, claims, invoices	Proof and settlement facts	Trip completion, pricing

04

FOUR AXES, FOUR DIFFERENT JOBS

The tenant is the marketplace operator

A region is not a tenant, and a shipper or transporter is not automatically a tenant. They are organisations participating in a marketplace tenant through scoped memberships.



- THE FOUR SCOPES

 - Region: placement and residency
 - Tenant: NATS account wall
 - Organisation: shipper, transporter, operator or integrator
 - Context: business-unit naming scope
- THE ENGAGEMENT RULE

 - No organisation receives another organisation's credential
 - No client publishes an award, allocation or settlement it does not own
 - Shared services cross roles through explicit domain commands and events

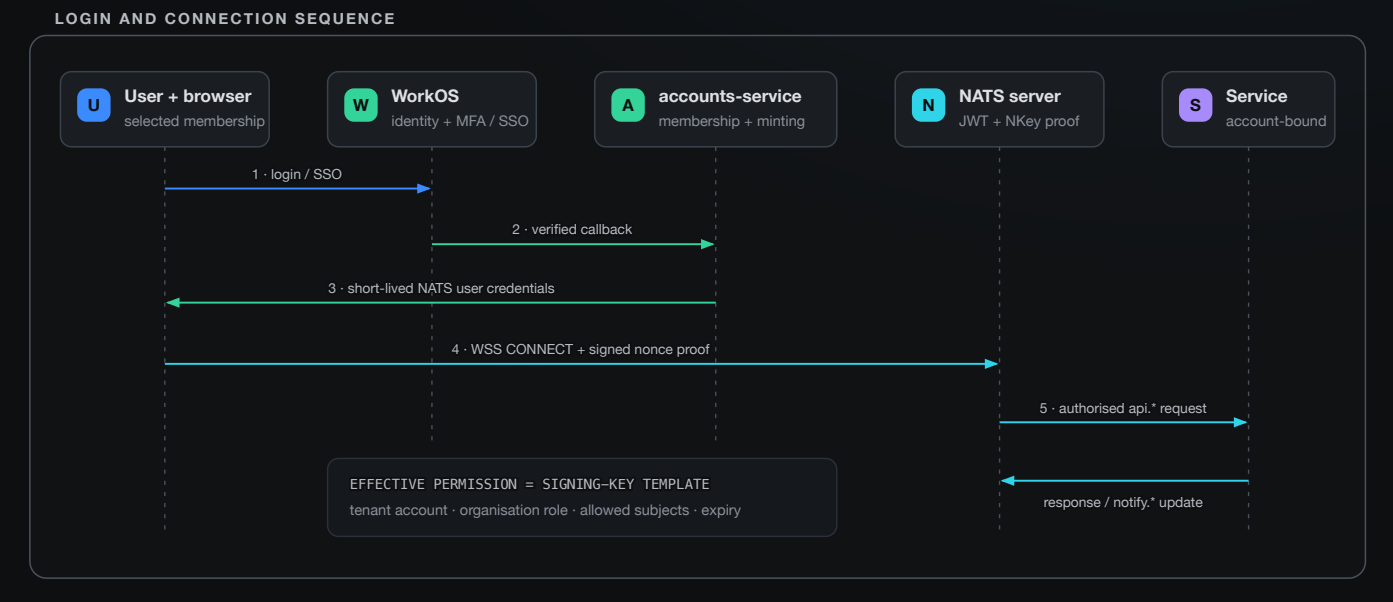
ACCOUNT	WHO CONNECTS	TYPICAL GRANT	NEVER GRANTED
PLATFORM	Auth, accounts, refdata, registry, observability	Platform <code>rpc.></code> and its own streams	Tenant business subjects
TENANT A	Tenant services and that tenant's browser users	<code>api.></code> plus <code>evt.></code> inside the account	Any subject in another tenant account
TENANT B	The same pattern, isolated namespace	Identical subject shapes	Cross-account subscribe
SYS / OBS	Operators and the observability service	Approved operational exports	Business payload subjects
Leaf / edge	Depot, warehouse, port and fleet gateways	A narrow publish set for that site	Broad wildcard subscribe

05

IDENTITY IS NOT MESSAGING AUTHORITY

From human sign-in to a NATS session

WorkOS establishes who the person is. Accounts-service decides which active membership they may use, then mints a short-lived credential whose signing key constrains the effective NATS permissions.



Blue = browser identity action · Green = identity and credential control · Cyan = authenticated NATS traffic · The browser never sends a tenant id that widens its credential

AUTHENTICATION

WorkOS handles password, enterprise SSO, MFA and directory-driven lifecycle.

AUTHORISATION

NATS validates the operator-account-user trust chain and enforces subject permissions server-side.

HTTP EXCEPTION

Where direct HTTP is unavoidable, first mint a short-lived, single-use capability over the authenticated NATS connection.

CREDENTIAL	ISSUED BY	LIFETIME	BOUND TO
WorkOS session	WorkOS	Provider policy	The human identity only
Membership selection	accounts-service	Per sign-in	One organisation membership in one tenant
NATS user JWT	accounts-service, scoped signing key	Short, renewed before expiry	One account and one least-privilege subject set
Service credential	Provisioning control plane	Long-lived and rotatable	One named process in one account
HTTP capability	The owning service, over NATS	Single use, seconds	One object or one operation

RENEWAL RULE

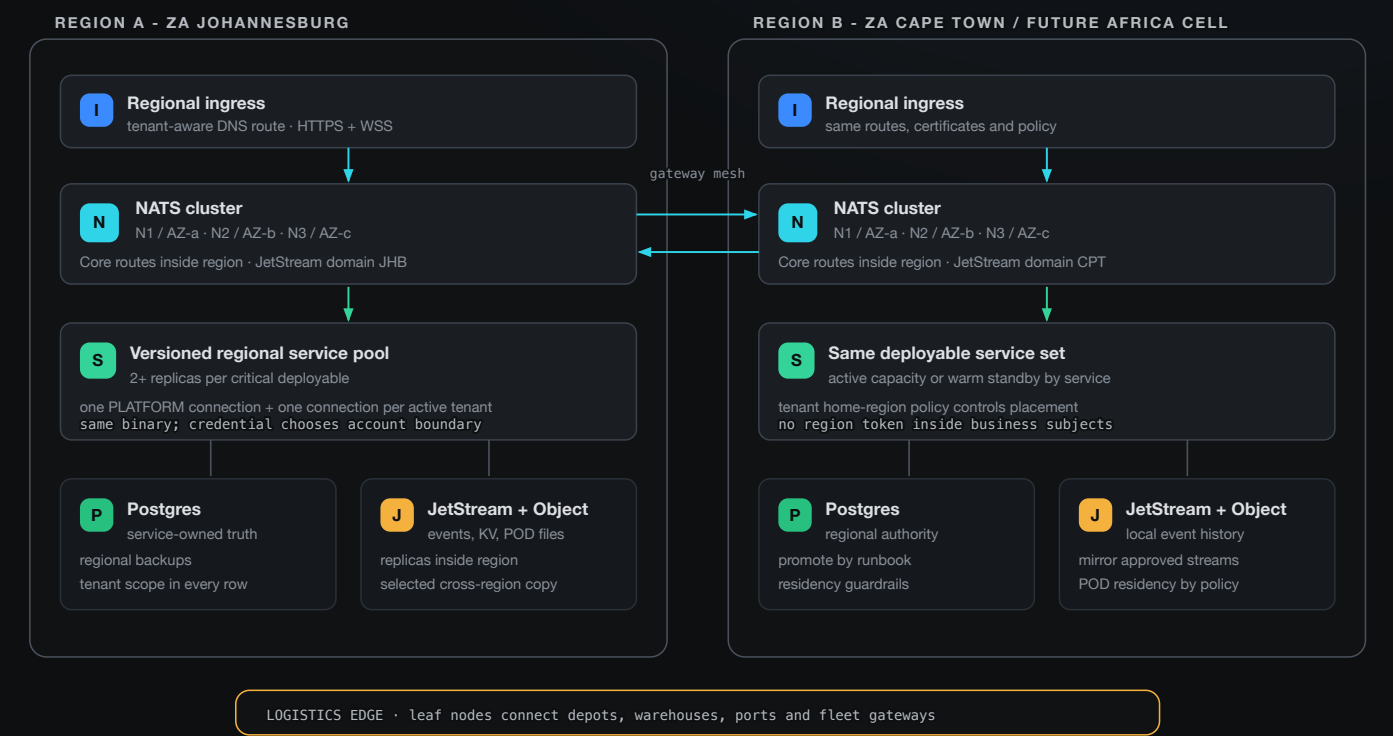
A session ends when its credential ends. The browser renews over the connection it already holds, and a revoked membership simply stops the next renewal — so withdrawing access never depends on a client honouring a logout.

06

REPEATABLE REGIONAL CELLS

Deploy regions, not bespoke tenant stacks

Each region is a repeatable cell: a three-node NATS cluster, the same versioned service set, service-owned persistence and regional ingress. Gateways connect cells without turning the WAN into one cluster.



- POOLED DEFAULT

 - One service deployment per region serves many tenant accounts
 - Each active tenant gets its own NATS connection and data scope
 - Best operational efficiency for the standard tier
- OPTIONAL SILO TIER

 - Dedicate selected service replicas or storage to a tenant
 - Keep the same subjects, contracts and onboarding control plane
 - Use for regulation, scale or noisy-neighbour isolation

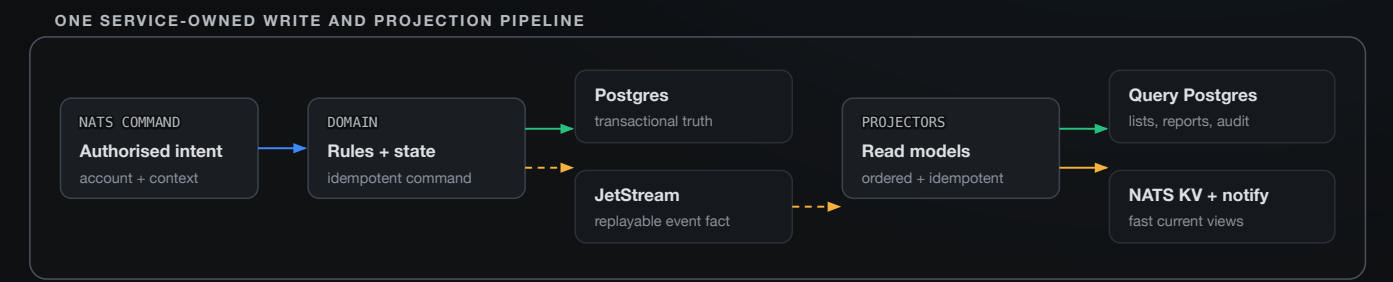
CONCERN	PER REGIONAL CELL	GLOBAL TO THE PLATFORM
NATS routing	A cluster of three nodes across availability zones	Gateways join the clusters into one supercluster
JetStream	A named domain holding that region's streams	Only approved mirrors may leave the region
Postgres	Service-owned authority plus regional backups	No global write master
Identity	Regional service endpoints and ingress	One directory and one NATS operator trust chain
Release	The same versioned service set, deployed locally	One catalogue of service and UI versions

07

TRUTH, HISTORY AND FAST VIEWS ARE DIFFERENT JOBS

Each service owns its facts

Postgres is authoritative for transactional state. JetStream retains replayable domain facts. KV and projections make current state fast to read; Object Store holds POD and compliance documents.



Blue = request · Green = authoritative write/read · Amber dashed = durable event/projection · Object Store follows the same ownership boundary for large immutable documents

DOMAIN	AUTHORITATIVE RECORDS	DURABLE EVENTS / OBJECTS	FAST VIEWS AND CONSUMERS
Marketplace	Tender, lane, offer, award, allocation	Published, bid submitted, awarded, allocated	Opportunity board, ranking, award audit
Fleet	Truck, trailer, driver, eligibility, availability	Vetted, availability changed, committed	Eligible-capacity index, transporter fleet view
Execution	Load, trip, route plan, milestone, exception	Dispatched, collected, in transit, delivered	Control tower, ETA, alerts, customer timeline
Tracking	Provider link, latest fix, consent and retention	Normalised position and exception events	Live map, ETA projection, corridor analytics
Documents	POD metadata, version, checksum, access policy	Immutable POD/compliance objects	POD retrieval, claims and settlement gate
Settlement	Charge, fee, claim, invoice, transporter payment	Approved, invoiced, paid, disputed	Statements, ageing, reconciliation and audit

EVENT-SOURCE WHEN

History is itself a business concern: replaying a trip, auditing an award, reconstructing allocation or proving a payment transition. Use CRUD when only current lookup state matters.

POSTGRES — TRUTH

Commits the decision in one transaction and answers what is contractually true right now.

JETSTREAM — HISTORY

Retains the fact under a limits policy, so a trip, award or payment sequence can be replayed later.

KV + OBJECT STORE

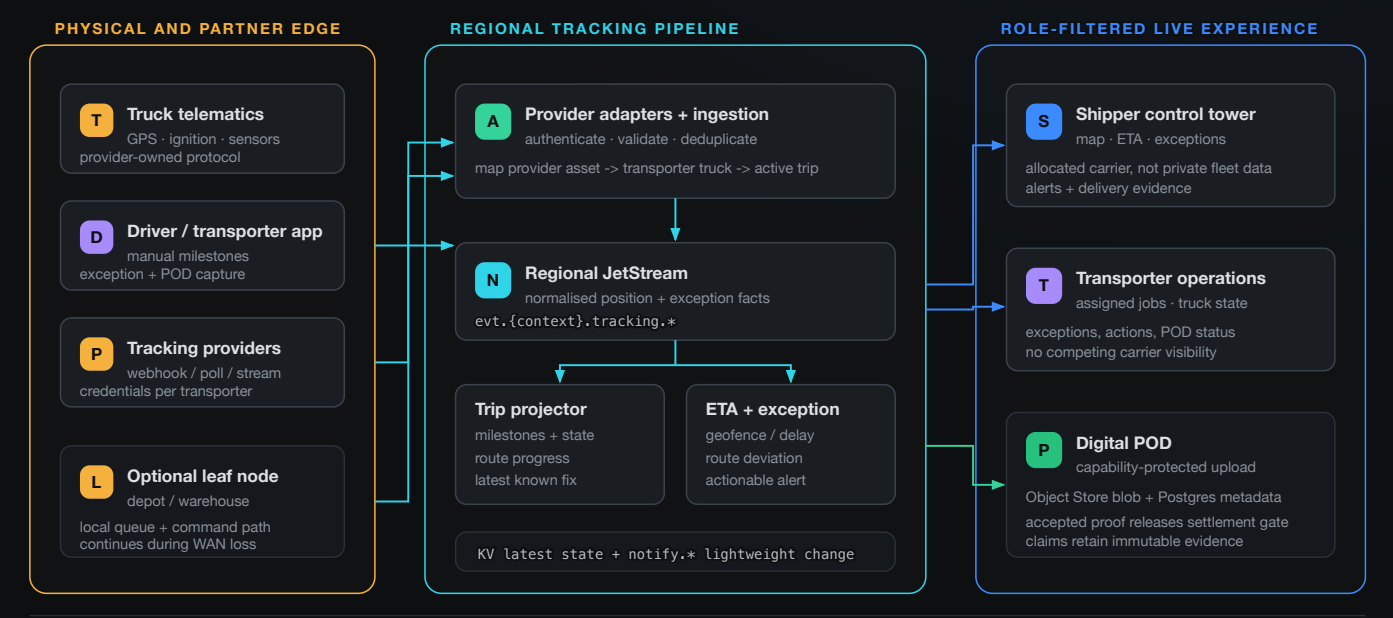
KV is an eager write-through cache for current views; Object Store holds POD blobs whose metadata stays in Postgres.

08

MANY TELEMETRY SHAPES, ONE TRIP TIMELINE

Tracking becomes a normalised business signal

Vehicle devices, transporter systems and driver apps differ. Provider adapters normalise their inputs before trip services derive milestones, ETA and exceptions for both client experiences.



Amber = external/edge source · Cyan = normalised event path · Blue = client projection · Green = accepted proof · Raw provider payloads never become the public domain contract

PRIVACY BOUNDARY

Share only tracking for an allocated load and its permitted time window. A shipper never gains general access to a transporter's fleet.

QUALITY BOUNDARY

Reject stale, impossible or duplicate fixes; retain provider provenance so ETA and dispute decisions remain explainable.

EDGE SOURCE	ARRIVES AS	ADAPTER MUST SUPPLY	DOMAIN RESULT
Vehicle telematics	Vendor position feed or webhook	Truck identity, timestamp, accuracy, provider	Normalised position fact
Transporter system	Batch or scheduled status file	Trip reference and milestone mapping	Milestone and exception fact
Driver app	Intermittent mobile updates	Ordering key and duplicate suppression	Trip event with device provenance
Depot / port gateway	Leaf-node publish from the site	Site identity and a narrow subject set	Gate-in, gate-out and dwell facts
Delivery evidence	Signature, photo or scanned document	Checksum, version and access policy	Immutable POD object plus metadata

ADAPTER RULE

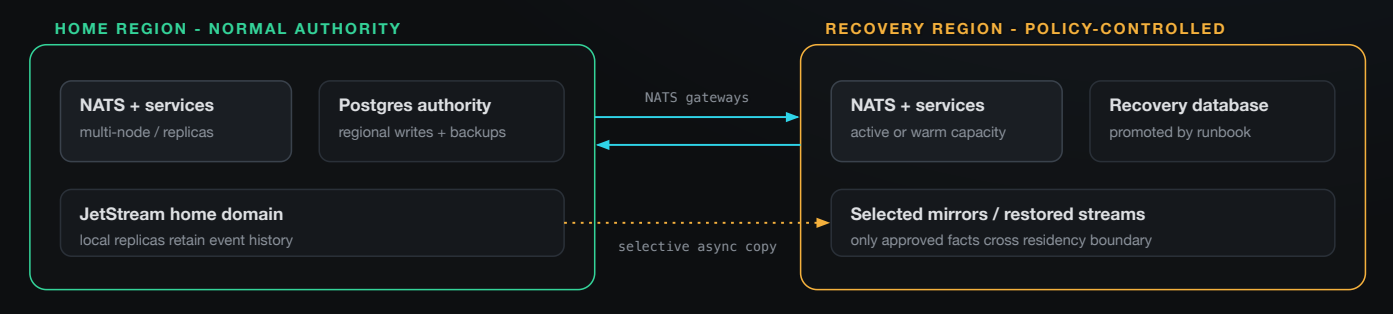
A raw provider payload is never the domain contract. If a provider changes its format, only its adapter changes; trip services, client projections and the settlement gate keep reading the same normalised events.

09

AVAILABILITY DOES NOT ERASE RESIDENCY

Fail locally before failing across regions

A node or service failure should be invisible inside its regional cell. A regional evacuation is a policy-driven event: traffic may move quickly, but authoritative data moves only through an approved DR design.



Cyan = live messaging topology · Amber dotted = asynchronous data movement · Blue = traffic steering · RTO and RPO are tenant policy inputs, not properties inferred from a diagram

FAILURE	EXPECTED RESPONSE	WHAT MUST NOT HAPPEN
One NATS node / AZ	Clients reconnect inside the cluster; JetStream leader changes	Cross-region failover for a local node loss
Service replica	Queue subscriber or load balancer selects a healthy replica	Duplicate non-idempotent command execution
WAN / gateway	Regions and leaf sites continue locally; replay after reconnection	Assume a disconnected edge is permanently failed
External GPS / WorkOS	Degrade the affected capability and surface stale/blocked state	Lose active trip state or widen access to compensate
Whole region	Apply tenant-specific routing, data-promotion and residency runbook	Promote unapproved or incomplete data globally

DECIDE CONTRACTUALLY

Home region, permitted recovery regions, RTO, RPO, POD residency and whether active trips may fail across borders.

TEST OPERATIONALLY

Node loss, account suspension, expired credentials, gateway partition, stale telemetry, projector replay and database promotion.

BLAST RADIUS

Contain a failure at the smallest boundary that can absorb it: node, then replica, then cell, then region.

EVIDENCE

Every failover records what moved, when it moved and which tenant policy permitted it.

RE-ENTRY

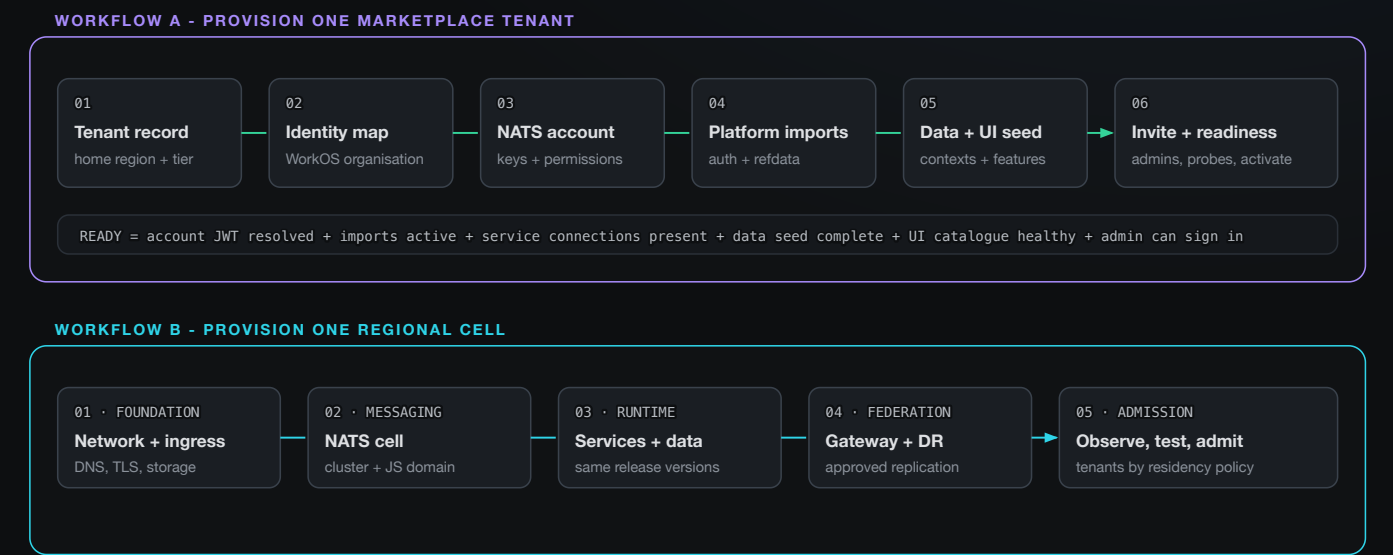
Returning to the home region is a planned operation with the same approval gates as leaving it.

10

TWO REPEATABLE CONTROL-PLANE WORKFLOWS

Add a tenant; clone a region

Neither operation is a ticket-driven collection of manual NATS edits. The control plane applies versioned policy, records every step and exposes readiness only when identity, messaging, data and UI agree.



DESIRED STATE

Tenant, account, imports, signing keys, regions, service release and UI catalogue are versioned configuration.

OBSERVED STATE

Connections, claims, consumers, stream health, projector lag, database readiness and plugin health are measured independently.

ACTIVATION GATE

Do not mark a tenant or region ready because provisioning returned 200. Require end-to-end probes from identity through a business read.

READINESS PROBE	WHAT IT PROVES	BLOCKS ACTIVATION WHEN
Identity round trip	A test member signs in and receives a scoped credential	Sign-in, membership or signing key is wrong
Account isolation	A cross-account publish is refused by the server	Imports or exports are wider than intended
Stream and consumer	A published fact is stored and then consumed	Domain, retention or consumer is missing
Projection read	A business read returns the projected fact	Projector lag or a migration is incomplete
UI catalogue	The shell loads the role-filtered contributions	A plugin manifest or version is unavailable

ROLLBACK RULE

Every provisioning step has a reverse step. Suspend an account, rotate or revoke a signing key, retire a release or withdraw a UI contribution — without deleting the tenant's history.

S

EVIDENCE AND DESIGN STATUS

What is sourced, settled and proposed

This document combines the repository's implemented NATS patterns, public statements about Linebooker's operating model and explicit architecture proposals for a future multi-region logistics SaaS.

REPOSITORY-GROUNDED

- › NATS operator mode with PLATFORM, tenant and system account boundaries
- › Accounts-service mints short-lived credentials; auth-service is merged into it
- › Per-tenant service connections; context-scoped subjects and KV keys
- › Postgres projection as truth with JetStream history and eager KV cache
- › WorkOS user lifecycle is documented, with implementation status called out separately

LINEBOOKER PUBLIC MODEL

- › Marketplace connects shippers/customers with vetted transport companies and trucks
- › Fixed contract lane rates dominate; bidding supports surplus capacity and return legs
- › Procure, aggregate and execute cover tendering, allocation, tracking, POD and payment
- › GPS-integrated tracking, digital POD, control-tower visibility and consolidated settlement
- › Transporters register, are vetted, receive opportunities, submit/accept, execute and get paid

ARCHITECTURE PROPOSALS REQUIRING DESIGN APPROVAL

- › Johannesburg and Cape Town shown as example regional cells; actual cloud regions and residency classes remain to be selected
- › Pooled regional services are the default, with an optional silo tier for regulation, scale or noisy-neighbour isolation
- › Tenant means marketplace-operating business; shipper and transporter organisations participate inside it through membership-scoped grants
- › Selected JetStream mirrors, Postgres recovery and Object Store replication are governed separately; no automatic global data movement is assumed
- › Domain boxes are bounded contexts; final service boundaries, event schemas, RTO/RPO and retention policies need business-rule approval

PRIMARY PUBLIC REFERENCES

Linebooker: [linebooker.com](#) · [Platform Solutions](#) · [For Transporters](#) · [FAQ](#)

NATS: [Adaptive Edge Architecture](#) · [Accounts](#) · [Gateways](#) · [Leaf Nodes](#)

AWS architecture framing: [SaaS Architecture Fundamentals](#) · [Architecting Multi-Region SaaS Solutions](#)

READING RULE

Public business statements describe outcomes, not internal implementation. Service names, NATS subjects, deployment tiers and failure behavior in these cards are the proposed technical model unless the repository already implements and documents them.